

Azure AI Search vs SQL Server 2025 Hyperscale

Native vector + semantic search comparison · cost crossover model · migration map · East US pricing, May 2026

Your workload

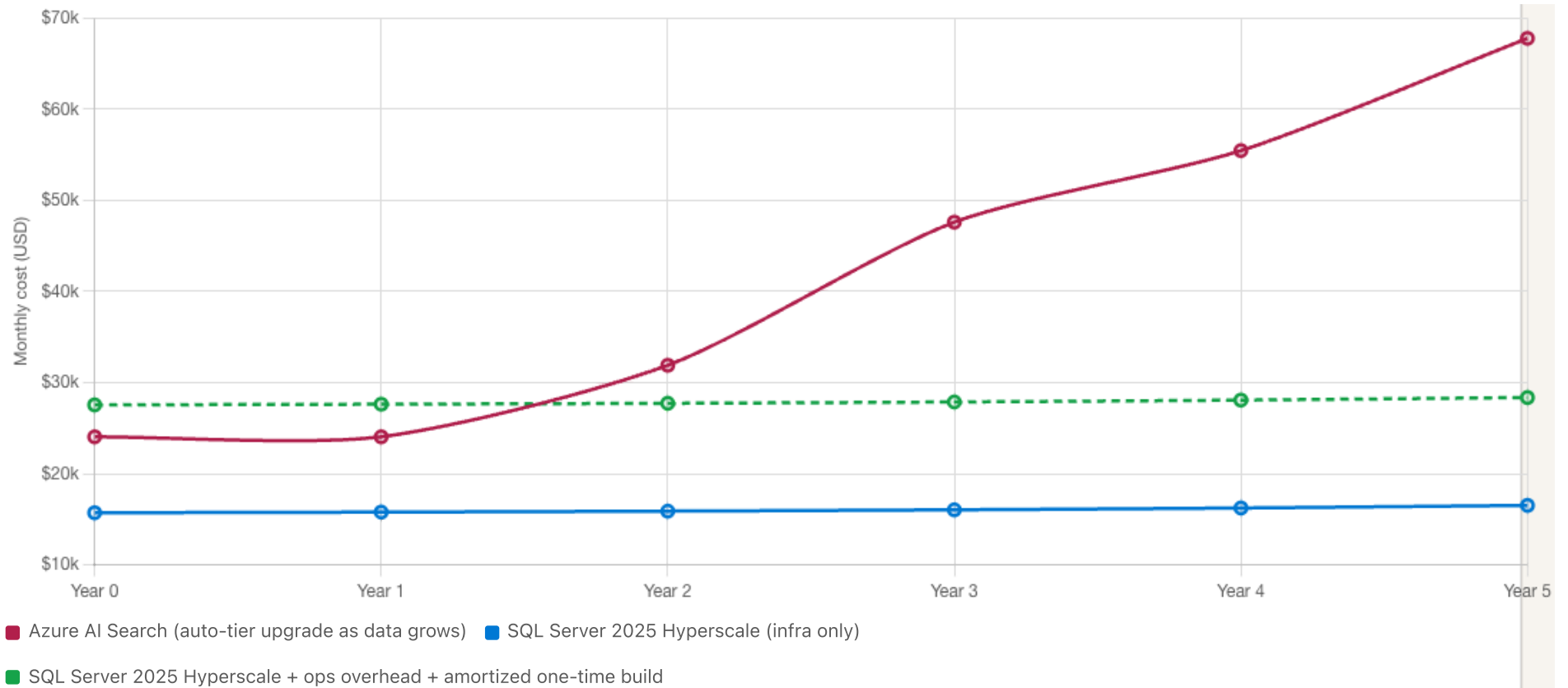
AI SEARCH BASELINE TIER	PARTITIONS (TODAY)	REPLICAS (TODAY)	YOY DATA GROWTH %	TIER STORAGE UTILIZATION %
Standard S2 (512 GB/par✓	6	4	40	60
SEMANTIC RANKER QUERIES / MO (K)	SQL VCORES PER REPLICA	SQL HA REPLICAS (INCL. PRIMARY)	SQL COMPUTE TIER	SQL RESERVATION
500	32	4	Premium Mem-Opt (\$0.2✓	1-year reserved (~35% ✓
SQL OPS OVERHEAD \$/MO (RECURRING)	SQL ONE-TIME BUILD COST \$ (NRE)	PROJECTION HORIZON (YEARS)		
6000	350000	5		

COST CROSSOVER

~ **Year 1.5**

Through year 1.5, **Azure AI Search is cheaper**. After that, SQL Hyperscale + ops overhead crosses below as AI Search is forced into higher tiers.

5-year cost projection



► Assumptions and how the model works

Year-by-year breakdown

YEAR	DATA (TB)	AI SEARCH TIER	PARTITIONS × REPLICAS	AI SEARCH \$/MO	SQL HS \$/MO	SQL HS ALL-IN \$/MO	CHEAPER
Year 0	1.80	Std S2	6 × 4 = 24 SU	\$24,047	\$15,716	\$27,549	AI Search
Year 1	2.52	Std S2	6 × 4 = 24 SU	\$24,047	\$15,789	\$27,623	AI Search
Year 2	3.53	Std S2	8 × 4 = 32 SU	\$31,896	\$15,893	\$27,726	SQL Hyperscale
Year 3	4.94	Std S3	6 × 4 = 24 SU	\$47,594	\$16,037	\$27,871	SQL Hyperscale

Year 4	6.91	Std S3	$7 \times 4 = 28$ SU	\$55,443	\$16,239	\$28,073	SQL Hyperscale
Year 5	9.68	Storage Opt L1	$6 \times 4 = 24$ SU	\$67,759	\$16,523	\$28,356	SQL Hyperscale